

INTRODUCTION TO DATA STRUCTURE

THE FOUNDATION OF
EFFICIENT PROGRAMMING!



EASY
EXPLANATION



BASIC
CONCEPTS



REAL LIFE
EXAMPLES



PERFECT
BEGINNER



ARRAY



LINKED LIST



STACK



QUEUE



TREE



GRAPH

What is Data?

- Data is raw facts and figures
- Can be numbers, text, images, etc.
- Example: 10, "infomax", 98.5

What is a Data Structure?

- A way of organizing and storing data
- Enables efficient access and modification
- Helps in better memory utilization

Need for Data Structures

- Efficient data management
- Faster access and processing
- Reduces time and space complexity
- Essential for large applications

Types of Data Structures

- **Primitive Data Structures**
 - int, float, char, pointer
- **Non-Primitive Data Structures**
 - Linear & Non-linear

- Elements arranged sequentially Easy to traverse Examples:
 - Array
 - Linked List
 - Stack
 - Queue

➤ Elements arranged hierarchically Complex relationships Examples:

➤ Tree

➤ Graph

Operations on Data Structures

- Insertion
- Deletion
- Traversal
- Searching
- Sorting

Characteristics of Data Structures



- Efficiency
- Reusability
- Abstraction
- Scalability

Efficiency of an Algorithm



- Algorithm efficiency measures performance
- Helps compare different algorithms
- Important for large-scale problems

What is an Algorithm?

- Step-by-step procedure to solve a problem
- Takes input and produces output

Why Efficiency Matters



- Saves time and resources
- Improves performance
- Essential for real-time systems

Types of Efficiency

- Time Efficiency
- Space Efficiency

Time Complexity

- Measures execution time of an algorithm
- Depends on input size (n)
- Expressed using Big-O notation

Common Time Complexities

- $O(1)$ – Constant time
- $O(\log n)$ – Logarithmic
- $O(n)$ – Linear
- $O(n \log n)$ – Linearithmic
- $O(n^2)$ – Quadratic

- Measures memory usage
- Includes:
 - Fixed part
 - Variable part

Asymptotic Notations

- Big-O (O) → Worst case
- Big-Omega (Ω) → Best case
- Big-Theta (Θ) → Average case

Factors Affecting Efficiency

- Input size
- Algorithm design
- Hardware & system environment

Thank You

Address: G. R. Complex Preetam Nagar, Dhoomanganj
Prayagraj Uttar Pradesh 211011

Mobile No.: 9598948810, 8874588766, 9532878816
Website : www.infomax.org.in

